

Discrete Math HW2

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1. *Geometric progression* — a sequence of non-zero numbers $b_1, b_2, b_3, \dots$, where each term is found by multiplying the previous one by constant number q : $b_{i+1} = b_i \cdot q$. To define a sequence it is sufficient to set the first element b_1 and the common ratio q .

Let $S_n = b_1 + b_2 + b_3 + \dots + b_n$, where $n \in \mathbb{N}$ and $b_{i+1} = b_i \cdot q$ with $q \neq 1$.

We can express each term in terms of the first element b_1 and the common ratio q :

$$S_n = b_1 + b_1q + b_1q^2 + \dots + b_1q^{n-1} = b_1(q^0 + q^1 + q^2 + \dots + q^{n-1})$$

To make the main statement hold, we will first prove by mathematical induction that for all $n \in \mathbb{N}$:

$$q^0 + q^1 + q^2 + \dots + q^{n-1} = \frac{1 - q^n}{1 - q}$$

Base Case ($n = 1$):

The left-hand side is $q^0 = 1$. The right-hand side is:

$$\frac{1 - q^1}{1 - q} = 1$$

Thus, the base case holds.

Inductive Step:

Assume that the statement holds for a specific integer $k \geq 1$:

$$q^0 + q^1 + \dots + q^{k-1} = \frac{1 - q^k}{1 - q}$$

Now, we check the statement for $k + 1$:

$$\begin{aligned}
 q^0 + q^1 + \dots + q^{k-1} + q^k &= \frac{1 - q^k}{1 - q} + q^k \\
 &= \frac{(1 - q^k) + q^k(1 - q)}{1 - q} \\
 &= \frac{1 - q^k + q^k - q^{k+1}}{1 - q} \\
 &= \frac{1 - q^{k+1}}{1 - q}
 \end{aligned}$$

Therefore, the inductive hypothesis holds for $k + 1$. By the principle of mathematical induction, the identity is true for all $n \in \mathbb{N}$.

Substituting this identity back into our equation for S_n :

$$S_n = b_1 \left(\frac{1 - q^n}{1 - q} \right)$$

Then the statement holds.

2. Prove that for every natural number n the equation holds true:

$$1^2 + 2^2 + \dots + n^2 = \frac{n(n+1)(2n+1)}{6}.$$

Consider base case $n = 1$.

Left part $= 1^2 = 1$

Right part $= \frac{1 \cdot 2 \cdot 3}{6} = 1$

Base case holds

Inductive hypothesis: suppose that for $k \geq 1$; $k \in \mathbb{N}$, the following statement holds:

$$1^2 + 2^2 + \dots + k^2 = \frac{k(k+1)(2k+1)}{6}$$

Then for $k + 1$:

$$\begin{aligned}
 1^2 + 2^2 + \dots + k^2 + (k+1)^2 &= \frac{k(k+1)(2k+1)}{6} + (k+1)^2 = \frac{(k+1)(2k^2+7k+6)}{6} = \\
 \frac{(k+1)(k+2)(2k+3)}{6} &= \frac{(k+1)((k+1)+1)(2(k+1)+1)}{6}
 \end{aligned}$$

Therefore the statement holds

3. Count the number of integers between 1 and 1000, that are divisible by 6 and 10

Consider the set $A = \{n \mid n \in N \text{ and } n \in (1, 1000) \text{ and } n = 6k, k \in N \text{ and } n = 10p, p \in N\}$

Since divisors 10 and 6 are not prime, perform prime number factorization to get LCM (6,10):

$$6 = 2 * 3$$

$$10 = 2 * 5$$

$$\text{LCM}(6,10) = 2 * 5 * 3 = 30$$

Then $A = \{n \mid n \in N \text{ and } n \in (1, 1000) \text{ and } n = 6k, k \in N \text{ and } n = 10p, p \in N\} = \{n \mid n \in N \text{ and } n \in (1, 1000) \text{ and } n = 30q, q \in N\}$

Every 30th number in sequence (1,1000) of natural numbers is divisible by 30.

$$\text{Then } \|A\| = \lfloor \frac{1000}{30} \rfloor = 33$$

4. Count the number of integers between 1 and 1000, that are divisible by either 3 or 10, but are not divisible by 15.

Consider sets: $A = \{n \mid n \in \mathbb{N} \text{ and } 1 \leq n \leq 1000 \text{ and } n = 3k, k \in \mathbb{N}\}$

$B = \{n \mid n \in \mathbb{N} \text{ and } 1 \leq n \leq 1000 \text{ and } n = 10p, p \in \mathbb{N}\}$

$C = \{n \mid n \in \mathbb{N} \text{ and } 1 \leq n \leq 1000 \text{ and } n = 15q, q \in \mathbb{N}\}$

By computing the sizes of these sets:

$$\|A\| = \left\lfloor \frac{1000}{3} \right\rfloor = 333$$

$$\|B\| = \left\lfloor \frac{1000}{10} \right\rfloor = 100$$

$$\|C\| = \left\lfloor \frac{1000}{15} \right\rfloor = 66$$

Number that are divisible either by 3 or 10, but not 15 belong to set $X = (A \cup B) \setminus C$.

By the inclusion-exclusion principle for set subtraction:

$$\|X\| = \|A\| + \|B\| - \|A \cap B\| - \|A \cap C\| - \|B \cap C\| + \|A \cap B \cap C\|$$

Where the intersections are defined as:

$$\begin{aligned}\|A \cap B\| &= \left\lfloor \frac{1000}{\text{lcm}(3,10)} \right\rfloor = \left\lfloor \frac{1000}{30} \right\rfloor = 33 \\ \|A \cap C\| &= \left\lfloor \frac{1000}{\text{lcm}(3,15)} \right\rfloor = \left\lfloor \frac{1000}{15} \right\rfloor = 66 \\ \|B \cap C\| &= \left\lfloor \frac{1000}{\text{lcm}(10,15)} \right\rfloor = \left\lfloor \frac{1000}{30} \right\rfloor = 33 \\ \|A \cap B \cap C\| &= \left\lfloor \frac{1000}{\text{lcm}(3,10,15)} \right\rfloor = \left\lfloor \frac{1000}{30} \right\rfloor = 33\end{aligned}$$

Substituting the values into the formula:

$$\begin{aligned}\|X\| &= 333 + 100 - 33 - 66 - 33 + 33 \\ \|X\| &= 334\end{aligned}$$

5. Count the number of integers between 100 and 999, that have at least one digit divisible by 4 (0 is divisible by any integer).

Suppose U is set of all possible numbers between 100-999 (900 numbers total). Suppose set A contains numbers with at least 1 digit divisible by 4. Suppose set B contains number with no digits divisible by 4.

Since no number can belong to A and B simultaneously, $|A| = |U| - |B|$

Each possible digit for $x \in B$ can be found in set $\{1, 2, 3, 5, 6, 7, 9\}$, 7 digits total. Then $|B| = 7^3 = 343$

Then $|A| = 900 - 343 = 557$

6. Prove de Morgan's laws:

$$\begin{aligned}\overline{A \cap B} &= \overline{A} \cup \overline{B} \\ \overline{A \cup B} &= \overline{A} \cap \overline{B}\end{aligned}$$

Using direct proof and logical de Morgan's laws:

$$(A \cap B)^c = \{x|x \notin (A \cap B)\} = \{x|x \notin A \vee x \notin B\} = \{x|x \in A^c \vee x \in B^c\} = A^c \cup B^c. \text{ Statement 1 is proven}$$

$$(A \cup B)^c = \{x|x \notin (A \cup B)\} = \{x|x \notin A \wedge x \notin B\} = \{x|x \in A^c \wedge x \in B^c\} = A^c \cap B^c. \text{ Statement 2 is proven}$$

7. Simplify the following expression using set operations and de Morgan's laws in such a way, that the complement is only applied to A, B or C, but not to any complex expression. Simplify the expression.

$$\neg(A \cup (\neg B \cap A) \setminus (\neg C \cup \neg(B \cup A)))$$

Left from backslash:

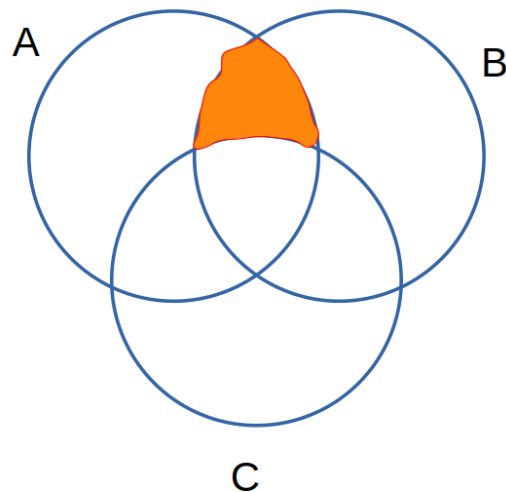
$$(A \cup \neg B) \cap A = A$$

Right from backslash:

$$(\neg C \cup \neg(B \cup A)) = \neg C \cup (\neg B \cap \neg A)$$

$$\text{Then } \neg(A \cup (\neg B \cap A) \setminus (\neg C \cup \neg(B \cup A))) = \neg(A \setminus (\neg C \cup (\neg B \cap \neg A))) = \neg((A \cap C) \cup (A \cap B \cap C)) = \neg(A \cap C) = \neg A \cup \neg C$$

8. Draw the Venn diagram for the set $\neg A \cup \neg B \cup C$



The requested set is **ALL EXCEPT** orange compartment

9. For each of the following binary relations on the set $A = \{a, b, c\}$ check, whether it is reflexive, anti-reflexive, symmetric, anti-symmetric, transitive:
1. $R_1 = \{(a, a), (a, b), (b, a), (b, b), (c, c)\}$
Reflexive, symmetric, transitive
 2. $R_2 = \{(a, a), (a, c), (c, b), (a, b), (b, b), (c, c)\}$
Reflexive, anti symmetric, transitive
 3. $R_3 = \{(a, a), (a, c), (c, b), (a, b)\}$
Anti-symmetric, transitive

10. Consider two sequences of sets A_i, B_i , where i ranges from 1 to n . Define $X = \bigcup_{i=1}^n A_i = A_1 \cup A_2 \cup \dots \cup A_{n-1} \cup A_n$, $Y = \bigcup_{i=1}^n B_i$, $Z = \bigcup_{i=1}^n (A_i \setminus B_i)$. Come up with an example where $X \setminus Y \neq Z$. Is any of inclusions $X \setminus Y \subset Z$ or $Z \subset X \setminus Y$ true? *P.S. To prove falsity, it is enough to provide a counterexample. To prove truth, a formal proof is required.*

1. Come with example where $X \setminus Y \neq Z$

Suppose $n = 2$; $A_1 = \{x\}$; $A_2 = \emptyset$; $B_1 = \emptyset$; $B_2 = \{x\}$

Then $X = \{x\} \cup \emptyset = \{x\}$

$Y = \emptyset \cup \{x\} = \{x\}$

$Z = (\{x\} \setminus \emptyset) \cup (\emptyset \setminus \{x\}) = \{x\}$

$X \setminus Y = \{x\} \setminus \{x\} = \emptyset \neq Z \longrightarrow$ counterexample found

2. Is inclusion $X \setminus Y \subset Z$ true?

Let $y \in X \setminus Y$. Then $(y \in X) \wedge (y \notin Y)$. Then $(y \in A_p, p \in [1, n]) \wedge (y \notin B_p, p \in [1, n])$

Since y can belong to any A_p , and y doesn't belong to any B_p , $y \in A_p \setminus B_p$.

Since Z is union of $A_i \setminus B_i, i \in [1, n]$, $y \in Z$. Then $y \in X \setminus Y$ also belong to Z , therefore inclusion $X \setminus Y \subset Z$ is true

3. Is inclusion $Z \subset X \setminus Y$ true?

This inclusion is not always true, see paragraph 1 of this task